

Creating a Multi-Sensory Horror Experience with VR

Danielle Cole

coles-danielle@shaw.ca
Carleton University
Ottawa, Ontario, Canada

Dexter Hunter Laroche

dhunterlaroche@gmail.com
Carleton University
Ottawa, Ontario, Canada

Mackenzie Darling

mackenziedarling4344@gmail.com
Carleton University
Ottawa, Ontario, Canada

Samuelle Lamoureux

samuelle.lamoureux@gmail.com
Carleton University
Ottawa, Ontario, Canada

Ryan Clement

ryanmpclement@yahoo.com
Carleton University
Ottawa, Ontario, Canada

Brendan O'Malley

brendanom99@gmail.com
Carleton University
Ottawa, Ontario, Canada

Lesley Istead

lesleyistead@gmail.com
Carleton University
Ottawa, Ontario, Canada

CCS CONCEPTS

• **Applied computing** → **Media arts**; • **Computing methodologies** → **Virtual reality**.

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1 INTRODUCTION

Terror arises not just from what we see, but what we hear and feel as well. In the dark of night, the sounds of a house settling can be terrifying, while objects that are slimy may illicit feelings of disgust. Our goal was to create a multi-sensory horror experience in virtual reality (VR) which combined digital elements with haunted house concepts—matching a surreal or fantastical visual experience with real-world sensations to create a greater sense of immersion. Our experience utilizes not only visuals, but also spatial audio and tactile feedback to create anxiety, terror, surprise, fear, and disgust. Previous work indicates that haptic and multi-sensory feedback (smell, temperature, etc), can enhance VR experiences and improve user engagement [1]. Using physical props related to the VR experience can increase stress and anxiety and therefore realism and immersion as shown in previous experiments [2]. We build upon these ideas using texture, moisture, and temperature to create the desired negative emotional response.

2 APPROACH

We created our VR experience using Unity and the Meta Quest 2 for a high-quality visual experience and accurate, marker-free hand tracking. Our experience consists of three separate scenes, each with a different sensory focus. The three scenes are connected by an "atrium", that allows the participant to choose the order of their experiences.

The first experience is a pitch black environment, creating a sense of isolation and playing into the "fear-of-darkness" as the participant cannot see the unfolding story. Spatial audio is used instead for storytelling and to immerse the participant. To increase immersion, head motion enables the participant to turn towards the sounds they hear around them.

The second experience is visual, building up tension over several minutes and eventually giving participants a jump-scare. The environment is dimly-lit, with subtle sounds and animations that encourage the viewer to look around to find the "source" of these events. The events increase in intensity and frequency, putting the viewer on edge and assisting in the build up to the scare point. In our preliminary experiments, participants expressed audible surprise even when knowing in advance what was coming.

The third experience utilizes various forms of tactile feedback to enhance the experience. The participant is presented with a scene of a cadaver in which they are instructed to stick their hand into and "get it out"—eventually finding an "egg" that explodes covering the view with spiders. We use hand and object tracking to provide an accurate representation of physical actions within the experience. While the participant is interacting with the digital environment, assistants are ensuring the participant feels the experience—using warm, goo-covered materials, including ropes and a tennis ball, spraying water, and using feathers to simulate the cadaver, egg, explosion, and spiders. The scenario ends with the viewer being instructed to place their "lovely" hands on the table, where the not-so-dead cadaver proceeds to "smash" them—simulated with an intense noise and air pressure. The realistic tactile feedback plays into common fears of blood/dissection, spiders, and having bones broken by providing harmless sensations that match visual elements.

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Figure 1: Scenes from our multi-sensory horror experience.

3 RESULTS

Our multi-sensory horror experience was presented at the Interactive Multimedia and Design senior project fair at Carleton University. To create a greater sense of immersion, the VR studio and assistants were decorated and costumed to match the storyline. A total of 41 consenting individuals tried our immersive VR experience. Three responded with loud screams, and two were unable to complete it (due to fear). Approximately 25% yelled or expressed surprise audibly. Overall, participants found the experience surprisingly terrifying, commenting that the audio-only experience was scarier than anticipated and that the tactile experience was disgusting, expressing a desire to "never do that again"—which was the desired response. While participants responded favourably,

with terror, fear, and disgust, to the experience, in future iterations we would like to integrate other sensory feedback such as smell to enhance the negative emotions created.

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